

FILIPPO CASTELLANI

PhD Candidate in Neuroengineering

...outside the lab, I improvise jazz lines with my band 🎵

[Portfolio Website](#)

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SUMMARY

I'm a 2nd year PhD student working on **invasive closed-loop neuroprosthetics for bladder control restoration**. My research combines the use of intraneural interfaces, neuromodulation techniques, neural decoding algorithms, and translational validation in the swine model.

I've worked on **Brain-Computer Interfaces clinical trials** for upper limb rehabilitation in post-stroke patients.

When abroad for my Master's thesis, in France, I studied **retinal visual information processing** at the level of single neurons and how these encode colors.

TECHNICAL EXPERTISE

Coding Languages: Python, Matlab, C++, JS, SQL, HTML, CSS.

Computational Modeling: Peripheral nerves hybrid models
• COMSOL: FEM Modeling
• NEURON: MRG Axons modeling

Electro-Physiology: EEG & EMG acquisition, FES stimulation, Neural recording & stimulation Spike sorting.

Machine-Learning Deep Neural Networks: [xR-AI project](#)
Decision Trees [CaBoLo project](#)

Others... Visit my website for past & ongoing [projects](#)

EXPERIENCE

10/2024 – 12/2025 **PhD @ Translational Neural Engineering Laboratory** Scuola Superiore Sant'Anna, Pisa, IT
Neuroprosthetics applications for Bladder Control Restoration. Bidirectional neural interfaces for Pudendal Nerve recording and stimulation. Supervisor: Prof. Silvestro Micera.

10/2023 – 04/2024 **Research internship (Master Thesis)** Institut de La Vision (Sorbonne Université), Paris, FR
Investigate visual information encoding in retinal neural circuits of the mouse.
• Design and execution of neural recording experiments using Multi Electrode Arrays.
• Analyze the activity of Retinal Ganglion Cells neural population in response to colored visual stimuli.

11/2021 – 09/2023 **Neurotechnology Researcher** S.Lucia Foundation IRCCS, Rome, IT
RECOMMENCER Project: *clinical trial - NCT05511207*
Research closed-loop Brain Computer Interfaces for upper-limb rehabilitation in post-stroke subjects.
• Implement real-time algorithms for EEG + EMG analysis, feature extraction and motion intention decoding for neurofeedback relay through Functional Electrical Stimulation (FES).
• Develop physiotherapist graphical interface for rehab sessions, software versioning and documentation.

PUBLICATIONS

2025 **Swine pudendal nerve intraneural stimulation enables selective neuromodulation of urethral and anal external sphincters.**
Castellani F.; Giannotti A.; Ceylan N.; Di Franco C.; Gabisonia K.; Musco S.; Del Popolo G.; Briganti A.; Miragliotta V.; Micera S.
Conference: [IEEE NER 2025](#) - BEST POSTER AWARD [\[Link\]](#)

2025 **Balanced spatiotemporal color responses are fine-tuned to natural light spectrum in mice retina**
Quétu T.[†], Louboutin A.[†], Castellani F.[†], Baroux R.; Ferrari U.; Goldin M.A.
Journal: [Vision Research 2025](#).

2022 **Cortico-Muscular Coupling to Control a Hybrid Brain-Computer Interface for Upper Limb Motor Rehabilitation: A Pseudo-Online Study on Stroke Patients.**
De Seta, V.; Toppi, J.; Colamarino, E.; Molle, R.; Castellani, F.; Cincotti, F.; Mattia, D.; Pichiorri, F.
Journal: [Frontiers in Human Neuroscience 2022](#).

EDUCATION

9/2021 – 7/2024 **MSc Biomedical Engineering** Politecnico di Milano, Milan, IT
scholarship holder Grade: 110/110 with Honors. Thesis: [\[Link\]](#)

9/2018 – 10/2021 **BSc Biomedical Engineering** Sapienza Università di Roma, Rome, IT
scholarship holder Grade: 110/110 with Honors. Publication: [\[Link\]](#)

LANGUAGES

English: B2 (Cambridge), **French:** C1 (Alliance Française), **Spanish:** B1 (Istituto Cervantes), **Italian:** native.